

Advanced Campus Area Network Design

Final Year Capstone Project

B.Tech — Electronics & Communication Engineering

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Abstract

This report presents the design, implementation, and validation of an advanced dual-campus university network developed as a final year capstone project. The work combines practical exposure gained during the internship at Scarlet Wireless India Pvt. Ltd. with the academic requirements of designing a secure, scalable, and highly available campus network for a university environment.

The proposed network follows a hierarchical three-tier architecture and uses VLAN segmentation, inter-VLAN routing, HSRP, LACP EtherChannel, OSPF dynamic routing, centralized wireless LAN controllers, DMZ-based server hosting, and a site-to-site IPsec VPN between two campuses. Cisco ASA firewalls enforce security policies, while access control, port security, and a Blackhole VLAN improve protection at the edge. Google Cloud integration demonstrates connectivity with external cloud services.

The entire network was implemented and tested in Cisco Packet Tracer 8.2.2. All 12 connectivity tests confirmed correct operation of DHCP, inter-VLAN communication, internet access, VPN tunneling, wireless access, HSRP failover, and secure SSH access.

Chapter 1 — Introduction

1.1 Background and Motivation

Modern universities are technology-intensive institutions. Thousands of students, faculty, and staff rely on the network every day for learning management systems, email, research databases, video conferencing, and cloud-based services. When the network fails or is compromised, academic operations stop. Designing a network that is simultaneously secure, highly available, scalable, and manageable is therefore one of the most critical challenges in institutional IT.

This project was motivated by practical experience gained during the internship at Scarlet Wireless India Pvt. Ltd., Mangaluru. Daily exposure to enterprise networking — including wireless deployments, firewall management, and user support — provided the real-world foundation for every design decision in this project.

1.2 Problem Statement

A university with two campuses serving 30,000 users across eight faculty departments requires a network that meets several demanding requirements simultaneously. Users need fast, reliable access to centralized services regardless of their location. The network must isolate different traffic types for security and performance. Branch campus users must securely access main campus servers without exposing those servers to the internet. All of this must remain available even when individual devices or links fail, and the architecture must scale to 60,000 users without redesign.

1.3 Objectives

- Design a dual-campus hierarchical network serving 30,000+ users, scalable to 60,000
- Implement VLAN segmentation for Management, LAN, WLAN, and Blackhole traffic
- Deploy Cisco ASA 5506-X firewalls with DMZ, Inside, and Outside security zones at each campus
- Configure OSPF dynamic routing across firewalls and multilayer switches
- Implement HSRP and LACP EtherChannel for gateway redundancy and link aggregation
- Deploy centralized WLC-based wireless with four SSIDs per department
- Configure site-to-site IPsec VPN between the two campuses
- Host DHCP, DNS, FTP, WEB, EMAIL, and SMTP servers in a protected DMZ
- Integrate with Google Cloud Platform for cloud service delivery

1.4 Scope

This project is designed and implemented entirely within Cisco Packet Tracer 8.2.2, which provides a realistic but simulated environment. Packet Tracer faithfully models OSPF, HSRP, LACP, VLANs, DHCP, and basic firewall behavior. Some advanced ASA features available on real hardware are identified as areas for future implementation.

Chapter 2 — Network Architecture

2.1 Hierarchical Three-Tier Model

The network follows Cisco's Hierarchical Three-Tier Model, which separates network functions into Core, Distribution, and Access layers. This separation provides modularity and fault isolation — failures at one layer are contained and do not propagate throughout the network.

Layer	Devices	Role
Core	Cisco ASA 5506-X (x2)	Perimeter firewall, IPsec VPN, ISP connectivity
Distribution	Catalyst 3850-24PS (x4)	Inter-VLAN routing, HSRP, OSPF, EtherChannel
Access	Catalyst 2960-24TT (x10)	End-device connectivity, VLAN assignment

2.2 Campus Structure

Main Campus — Five Faculty Departments:

- Health & Sciences
- Business
- Engineering
- Arts & Design
- Information Technology

Branch Campus — Four Faculty Departments:

- Health & Sciences
- Business
- Engineering
- Arts & Design

Each department has a dedicated Catalyst 2960-24TT access switch with PCs, printers, laptops, tablets, smartphones, and a CAPWAP-managed LAP.

Chapter 3 — IP Addressing Scheme

3.1 IP Address Allocation Summary

Network Zone	Campus	IP Range	Subnet Mask	Purpose
WLAN Management	Both	192.168.10.0/24	255.255.255.0	WLC & AP Management
LAN	Main	172.16.0.0/16	255.255.0.0	Main Campus wired LAN
LAN	Branch	172.17.0.0/16	255.255.0.0	Branch Campus wired LAN
WLAN	Main	10.10.0.0/16	255.255.0.0	Main Campus wireless
WLAN	Branch	10.11.0.0/16	255.255.0.0	Branch Campus wireless
DMZ Server Farm	Main	10.20.20.0/27	255.255.255.224	Servers (30 hosts)
Public WAN	Main	105.100.50.0/30	255.255.255.252	HQ ISP uplink
Public WAN	Branch	205.200.100.0/30	255.255.255.252	Branch ISP uplink
FWL-MLS1	Main	10.20.20.32/30	255.255.255.252	HQ FWL to MLSW1
FWL-MLS2	Main	10.20.20.36/30	255.255.255.252	HQ FWL to MLSW2
FWL-MLS1	Branch	10.20.20.40/30	255.255.255.252	BR FWL to MLSW1
FWL-MLS2	Branch	10.20.20.44/30	255.255.255.252	BR FWL to MLSW2
Google Cloud	Cloud	8.0.0.0/8	255.0.0.0	Google Cloud services

3.2 DMZ Server IP Assignments

IP Address	Device	Role
10.20.20.1	HQ-FWL (DMZ interface)	Firewall gateway for DMZ
10.20.20.5	DHCP-1 (Primary)	Primary DHCP server
10.20.20.6	DHCP-2 (Redundant)	Secondary DHCP server
10.20.20.7	DNS	Domain Name Resolution
10.20.20.8	WEB	University web portal
10.20.20.9	EMAIL	Email services
10.20.20.10	FTP	File transfer services

Chapter 4 — VLAN Architecture

VLAN segmentation logically divides the physical network into separate broadcast domains, confining broadcasts to members of the same VLAN regardless of physical switch location. This improves both security and performance.

VLAN ID	Name	Network	Purpose
10	MANAGEMENT	192.168.10.0/24	Network device management
20	LAN	172.16.0.0/16 (HQ) / 172.17.0.0/16 (BR)	Wired end-user devices
50	WLAN	10.10.0.0/16	Wireless clients — Main campus
90	WLAN	10.11.0.0/16	Wireless clients — Branch campus
199	BLACKHOLE	No routing	All unused switch ports — security only

All unused switch ports are assigned to VLAN 199 (Blackhole) and administratively shut down. This prevents any device plugged into an unused port from gaining network access.

Chapter 5 — Security Architecture

5.1 Firewall Zones (Cisco ASA 5506-X)

The Cisco ASA 5506-X is a stateful firewall that tracks the state of every connection and enforces policies based on zone-pair relationships. Three zones are configured at each campus:

Zone	Security Level	Purpose
Outside	0	Faces ISP — untrusted internet
DMZ	50	Hosts public-facing servers
Inside	100	Campus LAN — fully trusted

5.2 DMZ Architecture

The DMZ hosts services accessible from both inside the campus and the internet — web servers, email servers, DNS, and FTP. By placing these servers in the DMZ rather than on the inside network, a compromise of a DMZ server does not grant the attacker direct access to internal resources. The firewall enforces strict policies: outside traffic can only reach specific published services in the DMZ; DMZ servers cannot initiate connections to inside users.

5.3 Site-to-Site IPsec VPN

A site-to-site IPsec VPN creates an encrypted tunnel between the HQ firewall and the Branch firewall over the public internet. All traffic between the two campuses passes through this encrypted tunnel, preventing eavesdropping or tampering.

Parameter	Value
Phase 1 Encryption	AES-256
Phase 1 Hash	SHA
Authentication	Pre-shared key
DH Group	Group 2
Phase 2 Transform	ESP-AES-256 + SHA-HMAC
Mode	Tunnel
HQ Peer IP	105.100.50.2
Branch Peer IP	205.200.100.2

5.4 Access Control Hardening

- SSH v2 restricted via ACL to NET-SEC-PC only — no other device can remotely manage switches
- All unused switch ports assigned to VLAN 199 (Blackhole) and administratively shut down
- STP PortFast enabled on all access ports — immediate transition to forwarding state

- BPDUguard enabled — port enters err-disabled state if a BPDU is received from a rogue switch
- All devices hardened with enable secrets, password encryption, SSH-only VTY access, and login banners

Chapter 6 — Redundancy & High Availability

6.1 HSRP — Gateway Redundancy

Hot Standby Router Protocol (HSRP) provides default gateway redundancy by creating a virtual IP address shared between an Active and a Standby multilayer switch. If the Active switch fails, the Standby takes over within seconds — end devices experience no change in their default gateway configuration.

Campus	HSRP Active	HSRP Standby	Virtual IP
Main	HQ-MLSW1	HQ-MLSW2	Per-VLAN VIP
Branch	B-MLSW1	B-MLSW2	Per-VLAN VIP

6.2 LACP EtherChannel — Link Aggregation

Link Aggregation Control Protocol (LACP) bundles multiple physical GigabitEthernet links between the two distribution switches into a single logical channel, providing 3 Gbps aggregate bandwidth and link-level redundancy. If one physical link fails, traffic continues over the remaining links without any interruption.

Chapter 7 — Routing

7.1 OSPF Dynamic Routing

OSPF (Open Shortest Path First) is the Interior Gateway Protocol used throughout the network. Each OSPF router maintains a complete map of the network topology using the Link State Database (LSDB). Dijkstra's SPF algorithm calculates the best path to each destination. When a link fails, OSPF routers quickly detect the change, flood updated link-state advertisements, and reconverge on new paths — typically within 30 seconds.

All devices participate in OSPF Area 0 (the backbone area). OSPF neighbour authentication using SHA is configured on all OSPF-enabled interfaces to prevent unauthorized route injection.

7.2 Inter-VLAN Routing

Inter-VLAN routing is performed by the Catalyst 3850 multilayer switches using Switched Virtual Interfaces (SVIs). Each VLAN has a corresponding SVI that acts as the default gateway for devices in that VLAN. DHCP relay (ip helper-address) on each SVI forwards DHCP requests to the DMZ DHCP servers.

Chapter 8 — Wireless Infrastructure

Wireless infrastructure uses a centralized controller-based architecture. Lightweight Access Points (LAPs) communicate with a central Wireless LAN Controller (WLC) using the CAPWAP protocol. This centralizes configuration, RF management, and security policy across all access points.

Parameter	Main Campus	Branch Campus
WLC	HQ-WLC (10.10.0.15)	BR-WLC (10.11.0.15)
Protocol	CAPWAP	CAPWAP
LAPs	5 (1 per department)	4 (1 per department)
SSIDs per LAP	4	4
SSID Names	Employees, Corporate Users, External Auditors, Guest Users	Branch variants of same
WLAN Network	10.10.0.0/16	10.11.0.0/16

Chapter 9 — Device Inventory

Device	Model	Qty	Location	Role
HQ-FWL	Cisco ASA 5506-X	1	Main Campus	Perimeter firewall, DMZ, IPsec VPN
BRANCH-FWL	Cisco ASA 5506-X	1	Branch Campus	Perimeter firewall, IPsec VPN
HQ-MLSW1/2	Catalyst 3850-24PS	2	Main Campus	Distribution — HSRP, OSPF, EtherChannel
B-MLSW1/2	Catalyst 3850-24PS	2	Branch Campus	Distribution — HSRP, OSPF, EtherChannel
Faculty SW (HQ)	Catalyst 2960-24TT	5	Main Campus	Access — 1 per faculty
Faculty SW (BR)	Catalyst 2960-24TT	4	Branch Campus	Access — 1 per faculty
DMZ-SW	Catalyst 2960-24TT	1	Main Campus DMZ	Server farm switch
HQ-WLC	WLC-2504	1	Main Campus	Wireless LAN Controller
BR-WLC	WLC-2504	1	Branch Campus	Wireless LAN Controller
LAPs (HQ)	LAP-PT	5	Main Campus	1 per faculty department
LAPs (BR)	LAP-PT	4	Branch Campus	1 per faculty department
Servers	Server-PT	6	Main Campus DMZ	DHCP x2, DNS, Web, Email, FTP
ISP Routers	Cisco 2911	2	Internet	HQ-ISP and Branch-ISP
Internet Router	Cisco 2911	1	Internet	Core internet connectivity
Google VSW	Catalyst 2960-24TT	1	Cloud	Google Cloud virtual switch

Chapter 10 — Connectivity Test Results

All 12 connectivity tests were conducted live within Cisco Packet Tracer 8.2.2. Each test produced the expected result.

Test	Source	Destination	Expected	Result
DHCP Allocation	HS-PC1 (VLAN20)	DHCP 10.20.20.5	IP in correct range	PASS
Inter-VLAN	PC (VLAN20)	Printer (VLAN20)	Ping via SVI	PASS
PC to DMZ Web	BUS-PC1	10.20.20.8 (WEB)	HTTP via FWL	PASS
PC to Internet	ENG-PC1	8.8.8.8 (Google)	Ping via NAT	PASS
Branch to HQ DMZ	BENG-PC1	10.20.20.7 (DNS)	Via IPsec VPN	PASS
VPN Tunnel	HQ-FWL	BRANCH-FWL	IKE SA established	PASS
SSH Restriction	Non-admin PC	Switch VTY	Connection refused	PASS
SSH Allowed	NET-SEC-PC	HQ-MLSW1 VTY	Login success	PASS
WLC-LAP	BUS-LAP	HQ-WLC 10.10.0.15	CAPWAP connected	PASS
HSRP Failover	Shutdown MLSW1	Virtual IP	Traffic via MLSW2	PASS
Cloud Access	Any PC	Google Server 8.x.x.x	Via internet path	PASS
Branch Internet	BART-PC2	8.8.8.8	Via Branch-FWL NAT	PASS

Chapter 11 — Performance Metrics

Metric	Value
OSPF Convergence Time	< 30 seconds after topology change
HSRP Failover Time	< 10 seconds
DHCP Lease Time	24 hours (dual-server redundancy)
EtherChannel Bandwidth	3 Gbps aggregate (3x GigabitEthernet)
CAPWAP Association Time	< 30 seconds from startup
Active VLANs per Campus	5 (VLAN 10 / 20 / 50 or 90 / 199)
Max Supported Hosts (LAN)	65,534 per /16 subnet
Max Supported Hosts (WLAN)	65,534 per /16 subnet

Chapter 12 — Future Scope

Enhancement	Description
Cisco ISE (802.1X NAC)	Port-based authentication with dynamic VLAN assignment based on user identity and device posture
Firepower Threat Defense	Next-generation IPS, URL filtering, application visibility, and malware detection
SD-WAN	Intelligent path selection and application-aware routing replacing static IPsec VPN
IPv6 Dual-Stack	Future-proofing the addressing plan with OSPFv3 and DHCPv6
SNMP v3 / NetFlow	Centralized network monitoring, traffic analytics, and proactive fault detection

Conclusion

This project successfully designed and implemented a comprehensive, secure, and highly available enterprise campus network serving 30,000+ users across two campuses. All design requirements were met: hierarchical three-tier design, VLAN segmentation, Cisco ASA firewall zones, OSPF dynamic routing, HSRP redundancy, LACP EtherChannel, centralized WLC wireless infrastructure, DMZ server farm, site-to-site IPsec VPN, Google Cloud integration, and complete security hardening.

The internship at Scarlet Wireless India Pvt. Ltd. provided the real-world context that shaped every design decision in this project. The combination of practical internship experience and structured academic design has produced a strong foundation for a career in networking, cybersecurity, and enterprise IT.

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